

Other work has considered the interpretation of regressions using $\operatorname{arcsinh}(Y)$ or $\log(1 + Y)$ from the perspective of structural equations models, as opposed to the potential outcomes model considered here. This literature has reached diverging conclusions: For example, [Bellemare and Wichman \(2020\)](#) conclude that coefficients from $\operatorname{arcsinh}(Y)$ regressions have an interpretation as a semi-elasticity, while [Cohn et al. \(2022\)](#) conclude that these estimators are inconsistent and advocate for Poisson regression instead. [Thakral and Tô \(2023\)](#) show that the semi-elasticities implied by OLS regressions using $\operatorname{arcsinh}(Y)$ or $\log(1 + Y)$ are sensitive to scale; they recommend instead the use of power functions Y^k , which they show are the only transformations (besides log) for which the implied semi-elasticities for OLS regressions are scale-invariant. In [Appendix C](#), we show that these diverging conclusions stem from the fact that the structural equations considered in these papers implicitly impose different restrictions on the potential outcomes—some of which are incompatible with zero-valued outcomes—and consider different target causal parameters. This highlights the value of a potential outcomes framework such as ours, which makes transparent what causal parameters are identifiable and what properties they can have.

Finally, there is a long history in econometrics of explicitly modeling the intensive and extensive margins in settings with zero-valued outcomes, such as [Tobin \(1958\)](#) and [Heckman \(1979\)](#). Broadly speaking, these methods impose parametric structure on the joint distribution of the potential outcomes, which allows one to separate out the intensive and extensive margin effects of a treatment (see [Appendix C](#) for technical details). Of course, the parametric restrictions underlying these approaches may often be difficult to justify in practice, which perhaps has contributed to the growth in the use of log-like transformations in place of approaches that explicitly model the extensive margin. Our paper shows that the presence of an extensive margin should not simply be ignored by taking a log-like transformation. It also clarifies what parameters can be learned in such cases without imposing restrictions on the joint distribution of the potential outcomes.

1.1 Setup and notation

Let $D \in \{0, 1\}$ be a binary treatment and let $Y \in [0, \infty)$ be a weakly positively-valued outcome.⁵ We assume that $Y = DY(1) + (1 - D)Y(0)$, where $Y(1)$ and $Y(0)$ are respectively the potential outcomes under treatment and control. We suppose that in some (sub-)population of interest, $(Y(1), Y(0)) \sim P$ for some (unknown) joint distribution P . We denote the marginal distribution of $Y(d)$ under P by $P_{Y(d)}$ for $d = 0, 1$. We assume that neither $P_{Y(0)}$ nor $P_{Y(1)}$ is a degenerate distribution at zero.

2 Sensitivity to scaling for transformations that behave like $\log(Y)$

We first consider average treatment effects of the form $\theta = E_P[m(Y(1)) - m(Y(0))]$ for an increasing function m . We note that θ corresponds to the ATE among the (sub-)population indexed by P ;

⁵The $\operatorname{arcsinh}$ transformation is sometimes used in settings where Y can be negative. We impose that $Y \in [0, \infty)$, and thus do not consider this case. See [Appendix B.2](#) for extensions of our results to settings with continuous treatments.

if P refers to the sub-population of compliers for an instrument, for instance, then θ is the local average treatment effect (LATE), rather than the ATE in the full population. We are interested in how θ changes if we change the units of Y by a factor of a . That is, how does

$$\theta(a) = E_P[m(aY(1)) - m(aY(0))]$$

depend on a ? Setting $a = 100$, for example, might correspond with a change in units between dollars and cents. Of course, if Y is strictly positive and $m(y) = \log(y)$, then $\theta(a)$ is the ATE in logs and does not depend on the value of a .

We consider “log-like” functions $m(y)$ that are well-defined at zero but behave like $\log(y)$ for large values of y , in the sense that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$. This property is satisfied by $\log(1 + y)$ and $\operatorname{arcsinh}(y)$, for example. Our first main result shows that if the treatment affects the extensive margin, then $|\theta(a)|$ can be made to take any desired value through the appropriate choice of a .

Proposition 1. *Suppose that:*

1. *(The function m is continuous and increasing) $m : [0, \infty) \rightarrow \mathbb{R}$ is a continuous, weakly increasing function.*
2. *(The function m behaves like log for large values) $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$.*
3. *(Treatment affects the extensive margin) $P(Y(1) = 0) \neq P(Y(0) = 0)$.*
4. *(Finite expectations) $E_{P_{Y(d)}}[|\log(Y(d))| \mid Y(d) > 0] < \infty$ for $d = 0, 1$.⁶*

Then, for every $\theta^ \in (0, \infty)$, there exists an $a > 0$ such that $|\theta(a)| = \theta^*$. In particular, $\theta(a)$ is continuous with $\theta(a) \rightarrow 0$ as $a \rightarrow 0$ and $|\theta(a)| \rightarrow \infty$ as $a \rightarrow \infty$.*

[Proposition 1](#) casts serious doubt on the interpretation of ATEs for functions like $\log(1 + Y)$ or $\operatorname{arcsinh}(Y)$ as (approximate) average percentage effects. While a percent (or log point) is entirely invariant to the units of the outcome, [Proposition 1](#) shows that, in sharp contrast, the ATEs for these transformations are *arbitrarily* dependent on units.

2.1 Intuition for [Proposition 1](#)

Loosely speaking, the result in [Proposition 1](#) follows from the fact that a “percentage” treatment effect is not well-defined for individuals who have $Y(0) = 0$ but $Y(1) > 0$.⁷ Any ATE that is well-defined with zero-valued outcomes must implicitly determine how much weight to place on changes along the extensive margin relative to proportional changes along the intensive margin.

When $m(Y)$ behaves like $\log(Y)$ for large values of Y , the importance of the extensive margin is implicitly determined by the units of Y . For intuition, suppose that we re-scale the outcomes so

⁶This assumption simply ensures that $E_{P_{Y(d)}}[|m(aY(d))| \mid Y > 0]$ exists for all values of $a > 0$.

⁷See [Delius and Sterck \(2020\)](#) for an intuitive discussion of this difficulty in the context of the $\operatorname{arcsinh}(\cdot)$ transformation. They write, “the concept of elasticity itself does not make sense with zeros” (p. 21).

that the non-zero values of Y are very large. Then for an individual for whom treatment changes the outcome from zero to non-zero, the treatment effect will be very large, since $m(Y(1)) \gg m(Y(0)) = m(0)$. Extensive margin treatment effects thus have a large impact on the ATE when the values of Y are made large. By contrast, changing the units of Y does not change the importance of treatment effects along the intensive margin by much, since for $Y(1) > 0$ and $Y(0) > 0$, we have that $m(Y(1)) - m(Y(0)) \approx \log(Y(1)/Y(0))$, which does not depend on the units of the outcome.

To see the roles of the extensive and intensive margins more formally, for simplicity consider the case where $P(Y(1) = 0, Y(0) > 0) = 0$, so that, for example, everyone who has positive income without receiving a training also has positive income when receiving the training.⁸ Then, by the law of iterated expectations, we can write

$$\begin{aligned} E[m(aY(1)) - m(aY(0))] &= P(Y(1) > 0, Y(0) > 0) \underbrace{E_P[m(aY(1)) - m(aY(0)) \mid Y(1) > 0, Y(0) > 0]}_{\text{Intensive margin}} \\ &\quad + P(Y(1) > 0, Y(0) = 0) \underbrace{E_P[m(aY(1)) - m(0) \mid Y(1) > 0, Y(0) = 0]}_{\text{Extensive margin}}. \end{aligned}$$

When a is large, $m(ay) \approx \log(ay)$ for non-zero values of y , and thus the intensive margin effect in the previous display is approximately equal to $E_P[\log(Y(1)) - \log(Y(0)) \mid Y(1) > 0, Y(0) > 0]$, the treatment effect in logs for individuals with positive outcomes under both treatment and control. This, of course, does not depend on the scaling of the outcome. However, the extensive margin effect grows with a , since $m(aY(1)) \approx \log(a) + \log(Y(1))$ is increasing in a while $m(0)$ does not depend on a . Thus, as a grows large, the ATE for $m(aY)$ places more and more weight on the extensive margin effect of the treatment relative to the intensive margin. We can therefore make $|\theta(a)|$ arbitrarily large by sending $a \rightarrow \infty$. By contrast, if $a \approx 0$, then $m(aY(d)) \approx 0$ with very high probability, and thus the ATE for $m(aY)$ is approximately equal to 0.

It is worth emphasizing that the arbitrary scale-dependence described in [Proposition 1](#) exists whenever the treatment affects the probability that the outcome is zero, regardless of whether the extensive margin is of direct economic interest or not.⁹ In some settings, the presence of zeros may correspond to a discrete economic choice (e.g. not participating in the labor market), and thus may be of direct interest. In other settings—for example, if the outcome is a yearly count of publications which is sometimes zero for idiosyncratic reasons—the extensive margin may be a “nuisance” rather than a direct economic object of interest.¹⁰ The result in [Proposition 1](#) highlights that regardless of the source of the zeros, an ATE for a log-like transformation is not interpretable

⁸A similar argument goes through without this restriction, but then there are two extensive margins, one for individuals with $Y(1) > 0 = Y(0)$, and the other for those with $Y(0) > Y(1) = 0$.

⁹Without an extensive margin, ATEs for transformations $m(\cdot)$ defined at zero still exhibit scale-dependence, though perhaps not arbitrarily so. See [Section 3.1](#) below for further discussion.

¹⁰One setting where nuisance zeros may arise is when the observed outcome Y is actually a mis-measured version of the true economic object of interest. For example, publications Y may be a noisy measure of true researcher productivity $Y^* > 0$. One possible remedy in this setting is to model the measurement error to recover the treatment effect on Y^* rather than on Y . In a similar vein, [Gandhi et al. \(2023\)](#) models the measurement error in product shares in demand estimation, which are sometimes zero in finite samples.

as a percentage, since the presence of the extensive margin effect makes it arbitrarily dependent on the units. Indeed, a percentage effect is not well-defined for individuals moving from zero to non-zero outcomes. Whether the zeros correspond to a discrete economic choice or not will be relevant, however, when considering the choice of alternative target parameter, a topic we return to in [Section 4](#) below.

2.1.1 Intuition for the special case of $\log(1 + Y)$

We can also develop some intuition for [Proposition 1](#) by considering the special case where $m(y) = \log(1 + y)$. In that case, we have that

$$\theta(a) = E[\log(1 + aY(1)) - \log(1 + aY(0))] = E\left[\log\left(\frac{1 + aY(1)}{1 + aY(0)}\right)\right]. \quad (1)$$

Note that

$$\lim_{a \rightarrow \infty} \log\left(\frac{1 + aY(1)}{1 + aY(0)}\right) = \begin{cases} \log\left(\frac{Y(1)}{Y(0)}\right) & \text{if } Y(1) > 0, Y(0) > 0 \\ 0 & \text{if } Y(1) = 0, Y(0) = 0 \\ \infty & \text{if } Y(1) > 0, Y(0) = 0 \\ -\infty & \text{if } Y(1) = 0, Y(0) > 0. \end{cases}$$

We thus see that the term inside the expectation in (1) diverges to ∞ for individuals with $Y(1) > 0, Y(0) = 0$, and likewise diverges to $-\infty$ when $Y(1) = 0, Y(0) > 0$. If on average the extensive margin effect is positive, then there are more individuals for whom the limit is $+\infty$ rather than $-\infty$, and thus (under appropriate regularity conditions) the ATE diverges to ∞ . Analogously, if the extensive margin effect is negative, then the ATE diverges to $-\infty$. Hence, we see that the magnitude of the ATE for $\log(1 + aY)$ diverges as $a \rightarrow \infty$ when the average effect on the extensive margin is non-zero. By contrast, as $a \rightarrow 0$, $\log(1 + aY(d)) \rightarrow \log(1) = 0$ for both $d = 0$ and $d = 1$, and thus the treatment effect converges to 0. [Proposition 1](#) shows that this dependence on units occurs for *any* log-like transformation, not just $\log(1 + Y)$, and thus this issue cannot be fixed by choosing a different log-like transformation ($\log(c + Y)$, $\operatorname{arcsinh}(Y)$, $\operatorname{arcsinh}(\sqrt{Y})$, etc.)

2.2 Additional remarks and extensions

Remark 1 (ATEs for $\log(c + Y)$). In some settings, researchers consider the ATE for $\log(c + Y)$ and investigate sensitivity to the parameter c . Observe that $\log(1 + aY) = \log(a(1/a + Y)) = \log(a) + \log(1/a + Y)$, and thus the ATE for $\log(1 + aY)$ is equal to the ATE for $\log(1/a + Y)$. Hence, varying the constant term for $\log(c + Y)$ is equivalent to varying the scaling of the outcome when using $m(y) = \log(1 + y)$. [Proposition 1](#) thus implies that if treatment affects the extensive margin, one can obtain any desired magnitude for the ATE for $\log(c + Y)$ via the choice of c . In particular, the ATE for $\log(c + Y)$ grows large in magnitude as $c \rightarrow 0$, and small as $c \rightarrow \infty$.

Remark 2 (Finite changes in scaling). [Proposition 1](#) shows that any magnitude of $|\theta(a)|$ can be achieved via the appropriate choice of a . How much does $\theta(a)$ change for finite changes in the scaling